Brand Rule Lab

CodeNexus India 2026 — Inkloom AI | Round 2 Submission

Team: rohitlongesha5 **Team Leader:** Rohit Longesha **Project:** Brand Rule Lab — Pre-publication review with a claim-to-fact ledger **Live Deployed Application:** https://brand-rule-lab-ceyrdcrbd2aaxl6d5ndxs4.streamlit.app/ **GitHub Repository:** https://github.com/Rohuu-afk/Brand-Rule-Lab

---

1. Project Overview & Problem Alignment

The Real Problem (Inkloom AI Brief & Round 1 Evidence)

Growing consumer-facing brands publish content rapidly across distributed marketing teams, creative agencies, and generative AI tools. In high-scrutiny sectors such as Indian D2C Personal Care & Beauty, unsubstantiated claims, outdated facts, inconsistent terminology, and off-brand visuals lead to major credibility damage and regulatory friction (e.g., ASCI reports 90% of reviewed Beauty & Personal Care cases required modification; 82% of scrutinised digital ads were sponsored social media ads).

Primary research conducted in Round 1 revealed recurring approval friction:

- **Peer 1:** "Annoying part toh design ka hi rehta... approval ni milta sabse... senior deny krdete" (turnaround: 1–2 days; rejections due to late factual edits).
- **Peer 2:** "Last-minute changes — caption is good but date, venue, sponsor logo is wrong and whole post goes back for redo" (turnaround: 1–4 days).
- **Peer 3:** "Can we make it pop?" or "This doesn't feel right" without actionable specifics (turnaround: 24–48h social, 3–7 days high-stakes).

Brand Rule Lab Wedge

Instead of simply rejecting content or relying on vague chatbot reviews, **Brand Rule Lab links every approved promotional claim to a stable identifier (`stable_claim_id`) and an exact versioned evidence record (`fact_version_id`) in an immutable claim ledger.** When facts change or expire, dependent claims are automatically flagged for review (`REVIEW_REQUIRED` / `EXPIRED`).

---

2. Target Users & Workflow Roles

Brand Rule Lab establishes role-based governance:

- **Creators:** Upload drafts (PDF, DOCX, TXT, images), run pre-publication checks, inspect cited findings with exact character offsets, and accept, edit, or dismiss suggestions with intent-preserving corrections.
- **Reviewers:** Inspect the approval queue, review deterministic and AI findings, demand revisions, and make authoritative approval decisions. Reviewers also classify source-extracted guidelines through the **Rule Readiness Audit**.
- **Product / Quality Fact Owners:** Onboard source materials, manage approved facts, increment fact versions, set validity dates/markets, and expire evidence.

---

3. End-to-End Workflow (Draft → Check → Suggest → Approve → Ledger → Monitor)

1. **Brand Knowledge Onboarding:** Upload brand manuals, clinical studies, and product specification sheets (PDF, DOCX, TXT, XLSX, PNG, JPEG). Files are parsed locally (with Tesseract OCR when needed). AI extraction proposes source-quoted facts and guidelines with strict page/document provenance.
2. **Rule Readiness Gate:** Reviewers audit rule proposals into four distinct readiness classes:

- `CODE_CHECKABLE`: strictly verified against a whitelisted predicate (`term_list`, `regex`, `numeric_range`, `date_window`, `required_text`, `wcag_contrast`, `asset_hash`) with mandatory positive/negative self-tests. Only

passing, human-approved rules enter the active `brand_rules` engine.

- `AI_ASSISTED`: advisory contextual guidance; never silently converted into blocking code predicates.
- `HUMAN_JUDGEMENT`: subjective matters (e.g., brand aesthetics) routed directly to human reviewers.
- `UNCLEAR`: marked for clarification; blocked from enforcement until resolved.

1. **Draft Intake:** Creator submits copy or uploads documents/creatives, selecting product, target market (e.g., `IN`), channel (e.g., `instagram`), and optional campaign brief.
2. **Hybrid Check Pipeline:**

- **Python Deterministic Predicates:** evaluate terminology, regex constraints, date windows, required statements, image SHA-256 match, and WCAG AA 4.5:1 text contrast.
- **Source-Preserved Claim Extraction:** Python parses sentences and clause spans with exact character offsets. External AI selects candidate IDs without inventing spans or offsets.
- **Scoped Fact Retrieval & Evidence Ranking:** SQLite filters approved facts strictly by product and market. The AI ranks claims against candidates, classifying relationships into `SUPPORTS`, `CONTRADICTS`, `RELATED_NOT_EVIDENCE`, or `NO_MATCH`.
- **Advisory Tone & Campaign Alignment:** evaluates tone against voice guidelines (e.g., V-04 calm/credible tone vs. absolute promises) and campaign briefs.

1. **Intent-Preserving Suggestions:** Suggestions provide qualified rewrites (e.g., replacing "clears breakouts in 3 days" with "within 4 weeks" citing clinical study MS-2024-F112).
2. **Role-Gated Human Approval:** Reviewers approve, request changes, or escalate. Unlinked or contradictory claims cannot be approved without recorded overrides.
3. **Claim-to-Fact Ledger:** Approvals mint an immutable record with a 12-character hex `stable_claim_id`, normalized text SHA-256 hash, reviewer signature, timestamp, and linked `fact_version_id`.
4. **Drift Monitoring:** When a Fact Owner updates a fact to a newer version or a fact expires, an atomic database update immediately flags linked claims as `REVIEW_REQUIRED` or `EXPIRED`, while unrelated claims remain `CURRENT`.

---

4. Completed Features (Round 2 Deliverables)

- [x] **Deterministic Python Rule Engine:** Pure Python evaluation for `term_list` (banned/allowed terms), `regex`, `numeric_range`, `date_window`, `required_text`, `asset_hash`, and `wcag_contrast`.
- [x] **Rule Readiness Audit System:** Classification, source quote extraction, predicate compilation, automated positive/negative self-test harness, and SQLite history tracking.
- [x] **AI-Assisted Claim Extraction & Ranking:** Candidate offset preservation, product/market SQL isolation, JSON-schema validation, and explicit `NOT_OBSERVED` handling.
- [x] **Dual AI Provider Architecture:** Primary NVIDIA NIM (`nemotron-3-nano-omni-30b-a3b-reasoning`) with automatic graceful fallback to Google Gemini (`gemini-3.8-flash`) and offline resilience.
- [x] **Verifiable Claim Ledger:** `stable_claim_id` minting, immutable text hashes, fact-version lineage tracking, and atomic SQL drift cascade.
- [x] **Role-Based Workflows:** Separate Creator, Reviewer, and Fact Owner interfaces in Streamlit with immutable audit logging.
- [x] **Multi-format Ingestion:** Local parsing of PDF (PyMuPDF), DOCX (python-docx), XLSX (openpyxl), TXT, and image OCR (pytesseract & Pillow).
- [x] **Visual & Accessibility Checks:** WCAG AA 4.5:1 contrast calculation, alt-text presence verification, and exact SHA-256 asset verification.
- [x] **Synthetic Demonstration Suite:** Fully populated reference data for *Monsoon Beauty Co.* and *GlowUp* demonstrating V-04 (tone), T-12 (terminology), F-112 (clinical timeline), and offer expiration.
- [x] **Automated Test Suite:** 236 passing automated unit and integration tests (`236 passed, 0 failed`).
- [x] **Synthetic Evaluation Harness:** Reproducible test runner measuring turnaround, 60-claim matching, drift propagation, and prompt-injection safety.
- [x] **Production Packaging:** Dockerfile with multi-stage OCR dependencies and container health check; clean `.gitignore` and `.env.example`.

---

5. Technology & Implementation Details

Component	Technology	Rationale & Implementation Details
UI Framework	Streamlit 1.66.1	Fast, reactive multi-role interface with sidebar navigation, diff highlighting, and audit viewers.
Primary AI Provider	NVIDIA NIM API	OpenAI-compatible HTTPS API calling `nvidia/nemotron-3-nano-omni-30b-a3b-reasoning` with bounded timeouts.
Fallback AI Provider	Google Gemini (google-genai 2.34.0)	Automated fallback on network/quota failure to `gemini-3.8-flash` / `gemini-3.1-flash-lite`.
Local Parsing & OCR	PyMuPDF, python-docx, openpyxl, pytesseract	Local extraction of documents and image OCR without transmitting binary files to cloud LLMs.
Database & Ledger	SQLite 3 (WAL mode)	Strict foreign keys, ACID compliance, zero external infrastructure dependency, and deterministic SQL cascades.
Rule Engine	Custom Python Engine	High-throughput, deterministic execution of allowlisted predicates with regex caching and contrast calculations.
Contract Validation	jsonschema 4.23.0	Strict schema validation for all structured model outputs before ingestion.
Testing Harness	pytest 8.3.5	236 comprehensive unit and integration tests covering security, drift, and edge cases.

---

6. Progress Against the Round 1 Plan & Explanation of Changes

Plan Item (Round 1 Deck)	Status in Round 2	Details & Changes Made
End-to-End Workflow (Slides 1, 5, 8)	`COMPLETED`	Fully operational Draft → Check → Suggest → Approve → Ledger → Monitor lifecycle across Creator, Reviewer, and Fact Owner roles.
Checks in Action (Slide 6)	`COMPLETED`	Implemented V-04 (tone), T-12 (terminology), F-112 (claim vs. evidence duration), date windows, OCR text extraction, and WCAG AA contrast.
Fact Ledger & Drift (Slide 7)	`COMPLETED`	SQL drift cascade flags superseded fact versions (`REVIEW_REQUIRED`) and expired facts (`EXPIRED`) in 1 SQL update with 0 LLM calls, matching the slide specification.
Technical Stack (Slide 9)	`COMPLETED`	Python, Streamlit, SQLite, PyMuPDF, python-docx, Tesseract OCR.
Evaluation Plan (Slide 10)	`COMPLETED`	Reproducible evaluation suite verifying claim matching, drift cascade, and adversarial prompt-injection resistance.
Retrieval Architecture	`REFINED`	**Change from Plan:** Round 1 mentioned exploring Chroma vector index. We chose **canonical SQLite scoped queries** for eligible fact retrieval. This avoids dual-store sync anomalies, eliminates external persistence risks, and guarantees strict product/market isolation.
AI Provider Resiliency	`ENHANCED`	**Addition to Plan:** Integrated NVIDIA NIM as primary inference with seamless Google Gemini fallback and complete offline resilience (`NOT_OBSERVED` state instead of failure).

---

7. Responsible Design, Privacy & Prompt Safety (Slide 12 Alignment)

- **Human Authority:** AI never approves claims or auto-publishes content. Reviewers and Fact Owners retain exclusive approval authority in role-gated screens (demonstration roles, not authentication or cryptographic RBAC).
- **Privacy and Data Flow:** Uploaded files are parsed locally and raw files are never sent to an AI provider. After the user gives consent, claim extraction and tone assessment receive the extracted draft text, and fact ranking receives one claim plus eligible approved fact candidates. Without consent, only deterministic checks run and AI checks show `NOT_OBSERVED`.
- **Prompt Safety & Injection Defense:** Draft copy is treated as untrusted data. External AI has no database-write tools or execution permissions. Rule predicates run independently in Python; adversarial injection attempts cannot suppress deterministic rule failures.
- **Clear Operational Boundaries:** Coverage boundaries are explicit: the ledger tracks claims approved inside Brand Rule Lab; external untracked changes require manual hash verification. Internal states (`UNLINKED`, `NOT_OBSERVED`, `REVIEW_REQUIRED`) are strictly separated and never masked as compliant passes.
- **AI Assistance Disclosure (Rule 7 Compliance):** In accordance with Rule 7, significant code structure, test generation, schema design, and documentation assistance were provided by AI coding assistants (Google Antigravity, Claude, and Gemini). All architectural decisions, role workflows, deterministic rule compilation, and claim ledger validation were directed, reviewed, and verified by the team.

---

8. Verification, Test Results & Live Performance

- **Automated Test Suite:**

```
====================== 236 passed, 6 warnings ======================
```

100% test pass rate across 236 comprehensive test cases covering schema migrations, deterministic predicates, claim extraction, role-gated screens (demo roles), drift cascades, AI provider configurations, and security constraints.

- **Live Model Verification:**

Primary inference was verified with live calls to NVIDIA NIM (`nvidia/nemotron-3-nano-omni-30b-a3b-reasoning`), successfully generating schema-validated claim extraction, evidence ranking against approved clinical evidence (F-112), and tone analysis. Google Gemini (`gemini-3.8-flash` / `gemini-3.1-flash-lite`) is configured as an automated fallback adapter. Automated regression tests use isolated fixtures with mocked providers to avoid external network dependency.

- **Synthetic Evaluation Results:**
- Claim-to-fact matching: 60/60 synthetic deterministic cases passed.
- Fact-drift propagation: 100% of affected linked claims flagged upon fact version bump or expiration in synthetic tests.
- Adversarial safety test: 30/30 synthetic injection drafts using a mocked provider failed to suppress deterministic rule violations.
- *Note: These are internal reproducible fixture checks; they do not constitute independent external validation or live model performance metrics.*

---

9. Team Contributions

- **Team Name:** `rohitlongesha5`
- **Team Leader:** Rohit Longesha
- **Member A (Backend & Logic — Rohit Longesha):**
- Architecture and SQLite schema with WAL mode and migration sequence.
- Deterministic rule engine (`term_list`, `regex`, `numeric_range`, `date_window`, `wcag_contrast`, `asset_hash`).
- Scoped fact retrieval, AI provider integrations (NVIDIA NIM & Google Gemini fallback), and JSON-schema validation.
- Claim ledger, stable ID hashing, and SQL drift cascade propagation.
- Automated test suite (236 test cases) and synthetic evaluation harness.
- **Member B — Himanshu (UI, Workflows & QA):**
- Streamlit user interface across Creator, Reviewer, and Fact Owner workflows.
- Rule Readiness Audit interface, proposal review screens, and self-test runner UI.
- Visual metadata inspection, contrast previews, and diff visualization.
- Synthetic test corpus creation, manual walkthrough verification, and demo script preparation.

---

10. Three Major Improvements for Round 3 (Future Roadmap)

1. **Pre-Flight CMS & Social Scheduler Gatekeeper (Direct Publishing Integrations):**

Build bi-directional webhooks and API connectors for Shopify, WordPress, Webflow, and social schedulers (Buffer, Hootsuite). Before any scheduled post goes live, a pre-flight webhook queries Brand Rule Lab with the content's hash. If the associated `stable_claim_id` is missing, expired, or marked `REVIEW_REQUIRED` due to an upstream clinical fact update, the release is automatically intercepted and held for re-approval.

1. **Multimodal Packaging & Video Frame Inspection:**

Extend local ingestion beyond static images and documents to packaging renders and video creative. Integrate a lightweight local pipeline using Whisper for audio transcription and PyMuPDF/OpenCV for keyframe extraction, verifying spoken disclaimers, on-screen supers, and packaging ingredient claims against approved regulatory facts.

1. **Enterprise RBAC, Single Sign-On & Cryptographic Merkle Ledger Export:**

Upgrade from simulated demonstration roles to enterprise-grade authentication (OIDC/SAML, Okta, Google Workspace) with strict organization-level multi-tenancy. Implement a cryptographically verifiable Merkle-tree export of the claim ledger, allowing brands to provide auditors (such as ASCI or FTC regulatory bodies) with mathematically provable timestamps and fact-lineage evidence for any published asset.